

Theory of Computing:

8. Revision + Context-Free Languages - Part 2



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Outline :

- Revision + Midterm Exam Questions
- Pumping Lemma
 - Explanation
 - Examples
- Closure Properties

Midterm Exam Questions + Revision



- **DFA/NFA**

- Write reference words :
 - From the language
 - Not from the language
- Start with the easy case and build on top of it.
- Complete the trap states for the case of DFA

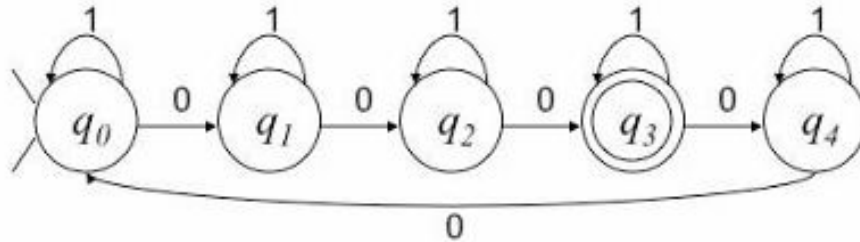
Midterm Exam Questions + Revision

- DFA/NFA
 - $\{ w \mid w \text{ has } n \text{ occurrences of } 0\text{'s where } n \bmod 5 \text{ is } 3. \}$

Midterm Exam Questions + Revision

- DFA/NFA

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Midterm Exam Questions + Revision

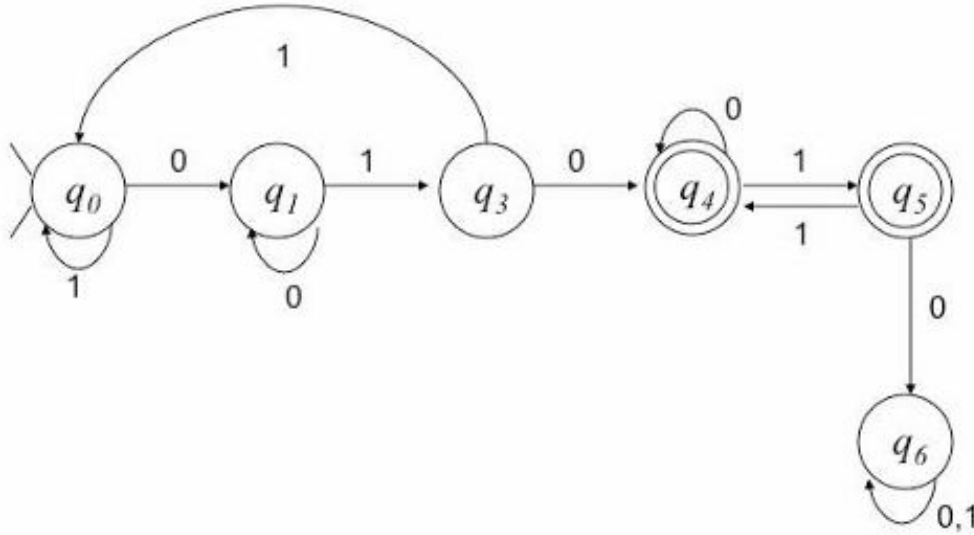


- DFA/NFA
 - $\{ w \mid w \text{ has exactly one occurrence of the substring } 010. \}$

Midterm Exam Questions + Revision

- DFA/NFA

- $\{ w \mid w \text{ has exactly one occurrence of the substring } 010. \}$



Midterm Exam Questions + Revision

A series of decorative horizontal bars in the top right corner: a pink bar, a dark grey bar, a blue bar, a yellow bar, and a grey hatched bar.

- **Regular Expressions**

- The language which does not contain the 10 as a substring. (From the lecture)

Midterm Exam Questions + Revision

- **Regular Expressions**

- The language which does not contain the 10 as a substring. (From the lecture)
 - 0^*1^*

Midterm Exam Questions + Revision

- **Regular Expressions**

- The language $\{w \in \Sigma^* \mid w \text{ contains exactly one double letter}\}$. For example, baaba has exactly one double letter, but baaaba has two double letters.

- $(\epsilon \mid b)(ab)^*aa(ba)^*(\epsilon \mid b) \mid (\epsilon \mid a)(ba)^*bb(ab)^*(\epsilon \mid a)$

Midterm Exam Questions + Revision

- Regular Expressions

- The language $\{w \in \Sigma^* \mid w \text{ does not end in a double letter}\}.$

- $\epsilon \mid a \mid b \mid (a|b)^*(ab|ba)$

Midterm Exam Questions + Revision



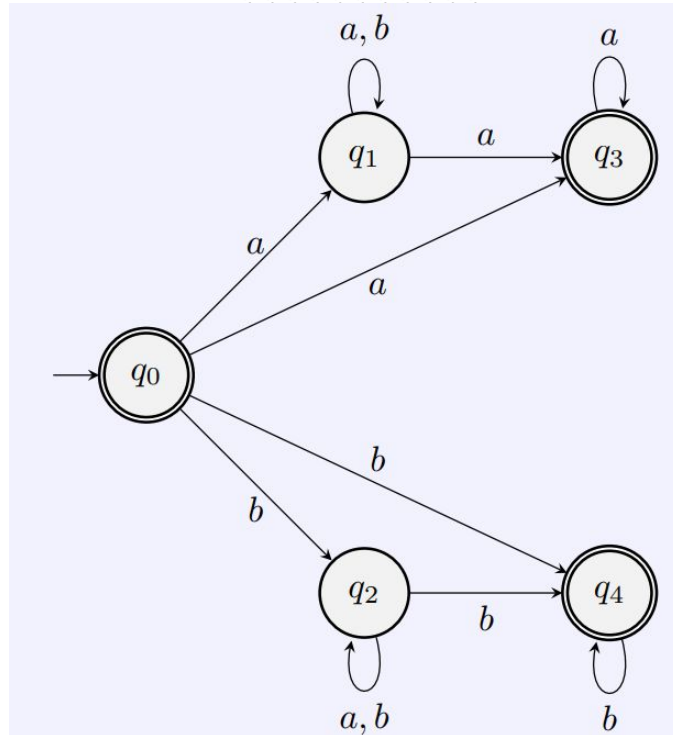
- **Regular Languages**

- $L = \{w \mid \text{the number of substring "ab" equals the number of substring "ba" in the word } w\}$ over the alphabet $\{a,b\}$

Midterm Exam Questions + Revision

- **Regular Languages**

- $L = \{w \mid \text{the number of substring "ab" equals the number of substring "ba" in the word } w\}$ over the alphabet $\{a,b\}$



Midterm Exam Questions + Revision

- Regular Languages

- $L = \{a^n b^m \mid n, m \geq 0, n > m\}$ over the alphabet $\{a, b\}$ [From Optional]

Midterm Exam Questions + Revision

- Regular Languages

- $L = \{a^n b^m \mid n, m \geq 0, n > m\}$ over the alphabet $\{a, b\}$ [From Optional]

- Pumping down : $s = xyz = a^{p+1}b^p$ for the case where $i=0$, $s' = xy^0z$ should be in the language, but $|y| > 0$, therefore, by pumping down, the number of a would be equal or less than the number of b . Which is not in the language.

Midterm Exam Questions + Revision

- **Regular Languages**

- $L = \{w \mid w = w^R \text{ and } |w| \text{ is odd, meaning that } w \text{ is a palindrome of odd length}\}$ over the alphabet $\{a\}$

Midterm Exam Questions + Revision

- Regular Languages

- $L = \{w \mid w = w^R \text{ and } |w| \text{ is odd}, \text{ meaning that } w \text{ is a palindrome of odd length}\}$ over the alphabet $\{a\}$
 - Words composed of a single letter are palindromes
 - Words of an odd length : **$a(aa)^*$**

Midterm Exam Questions + Revision

- **Regular Languages**

- $L = \{0^k \mid k \text{ is a prime number} \}$. Provided that the set of prime numbers is infinite.(no need to prove this).

Midterm Exam Questions + Revision

- Regular Languages

- $L = \{0^k \mid k \text{ is a prime number} \}$. Provided that the set of prime numbers is infinite.(no need to prove this).

- Let's choose the string $s=0^P$ such that P is a prime number and P is greater than the pumping length called L .
- Assume that $|y|=k$, Let's pump y , $P+1$ times, so that the word will be : $xy^{P+1}z$ The number of zeros will be $|xyz| + p|y| = p+pk$, therefore the generated word is 0^{p+pk}
- But $P+PK = P(1+K)$ is not a prime number, therefore, the newly generated word is not in the language → Contradiction to the assumed hypothesis that if you pump, you will always get words in the language. Therefore, the language is not regular.

Midterm Exam Questions + Revision

- Regular Languages

- $L = \{xww^R y \mid x, y \in \Sigma^* \text{ and } |w|, |x|, |y| \geq 1\}$

Midterm Exam Questions + Revision

- Regular Languages

- $L = \{xww^R y \mid x, y \in \Sigma^* \text{ and } |w|, |x|, |y| \geq 1\}$

- $(\Sigma^*00\Sigma^*) \mid (\Sigma^*11\Sigma^*)$ (There is x and y to absorb letters)

Midterm Exam Questions + Revision

- Regular Languages

- $L = \{xww^R \mid x, y \in \Sigma^* \text{ and } |w|, |x|, |y| \geq 1\}$

Midterm Exam Questions + Revision

- **Regular Languages**

- $L = \{xww^R \mid x, y \in \Sigma^* \text{ and } |w|, |x| \geq 1\}$
- No DFA, no RegEx, it is non-regular
 - $\Rightarrow$ use the pumping lemma:
 - $\Rightarrow s = 00^P 10^P \quad |xy| \leq P \dots$ Pump $Y \rightarrow$ always more zeros on the left side, but still the words will be in the language

0000 $0^P 10^P$

000000..000 $0^P 10^P$

Midterm Exam Questions + Revision

- Regular Languages

- $L = \{xww^R \mid x, w \in \Sigma^* \text{ and } |w|, |x| \geq 1\}$

Midterm Exam Questions + Revision



- **Context Free Grammars**
 - The language $\{a^n b^n \mid n \geq 0\}$ over the alphabet $\{a, b\}$

Midterm Exam Questions + Revision

- **Context Free Grammars**

- The language $\{a^n b^n \mid n \geq 0\}$ over the alphabet $\{a, b\}$

■ $S \rightarrow aSb \mid \epsilon$

Midterm Exam Questions + Revision

- **Context Free Grammars**

- The language $\{a^n b^m \mid n \neq m\}$ over the alphabet $\{a, b\}$

Midterm Exam Questions + Revision

- **Context Free Grammars**

- The language $\{a^n b^m \mid n \neq m\}$ over the alphabet $\{a, b\}$
 - $S \rightarrow aSb \mid U \mid T$
 - $U \rightarrow$
 - T

→

Midterm Exam Questions + Revision

- **Context Free Grammars**

- The language $\{a^n b^m \mid n \neq m\}$ over the alphabet $\{a, b\}$

- $S \rightarrow aSb \mid U \mid T$

- $U \rightarrow aU \mid a$

- $T \rightarrow Tb \mid b$

Midterm Exam Questions + Revision

- **Context Free Grammars**

- The **complement** of the language $\{a^n b^n \mid n \geq 0\}$ over the alphabet $\{a, b\}$
[From optional]

Midterm Exam Questions + Revision

- Context Free Grammars

- The **complement** of the language $\{a^n b^n \mid n \geq 0\}$ over the alphabet $\{a,b\}$ [From optional]

■ $S \rightarrow S_1 \mid S_2$
 S_1 should be $a^n b^m$ such that m is not equal to n
 S_2 words of the form : $\{a,b\}^* \mathbf{ba} \{a,b\}^*$

Midterm Exam Questions + Revision

- **Context Free Grammars**

- The complement of the language $\{a^n b^n \mid n \geq 0\}$ over the alphabet $\{a, b\}$ [From optional]

■	$S \rightarrow S_1 \mid S_2$		Y		Z
	$S_1 \rightarrow aS_1b \mid T \mid U$		be		b^*a^*
	$S_2 \rightarrow RbaR$				
	$T \rightarrow aT \mid a$				
	$U \rightarrow Ub \mid b$				
	$R \rightarrow RR \mid a \mid b \mid \epsilon$				

Midterm Exam Questions + Revision

- **Context Free Grammars**

- The language $L = \{x\#y \mid |x| \neq |y|\}$ over the alphabet $\{a, b, \#\}$. (Examples of words in the language : $aa\#bbb$, $aaaa\#b$.. Words not in the language : $a\#b$ }

Midterm Exam Questions + Revision

- **Context Free Grammars**

- The language $L = \{x\#y \mid |x| \neq |y|\}$ over the alphabet $\{a, b, \#\}$. (Examples of words in the language : $aa\#bbb$, $aaaa\#b$.. Words not in the language : $a\#b$ }

$$S \rightarrow XSX | XL | RX$$

$$L \rightarrow \# | XL$$

$$R \rightarrow \# | RX$$

$$X \rightarrow 0 | 1$$

Midterm Exam Questions + Revision



- **Context Free Grammars**

- Construct the Context-Free Grammar for the language where the number of b + 2 = the number of a . The number of a is more than b but with strictly only two letters.

Midterm Exam Questions + Revision



- **Context Free Grammars**

- Construct the Context-Free Grammar for the language where the number of b = the number of a .

Midterm Exam Questions + Revision

- **Context Free Grammars**

- Construct the Context-Free Grammar for the language where the number of b = the number of a.

- $S \rightarrow 0S1 \mid 1S0 \mid SS \mid \epsilon$

OR

- $S \rightarrow 0S1S \mid 1S0S \mid \epsilon$

Midterm Exam Questions + Revision

- **Context Free Grammars**

- Construct the Context-Free Grammar for the language where the number of b + 2 = the number of a . The number of a is more than b but with strictly only two letters.
 - $S \rightarrow aSb \mid bSa \mid SS \mid aa \mid ?$

Midterm Exam Questions + Revision

- **Context Free Grammars**

- Construct the Context-Free Grammar for the language where the number of b + 2 = the number of a . The number of a is more than b but with strictly only two letters.
 - $S \rightarrow aSb \mid bSa \mid SS \mid aa \ ?$
 - Can generate : $aaaa \ ?$

Mid + Re

Step 3: Final CFG

plaintext

Copy

Edit

Variables: S

Terminals: a, b

Start Symbol: S

Productions:

$S \rightarrow aa$

| aSb

| bSa

| Sa b

| bS a

ChatGPT

Note:

- You can simplify and just keep a few combinations like $S \rightarrow aSb \mid bSa \mid aa$, which is sufficient.
- This CFG generates all strings over $\{a, b\}$ where number of a's is **exactly 2 more** than number of b's.

Midterm Exam Questions + Revision

Gemini

Construct the Context-Free Grammar for the language where the number of $b + 2 =$ the number of a . The number of a is more than b but with strictly only two...



Alright, let's build that grammar! You're looking for strings made of only 'a's and 'b's where the count of 'a's is exactly two more than the count of 'b's, and there's at least one 'a' and one 'b'.

Here's a context-free grammar that nails it:

```
S -> aaB  
B -> bB | ε
```

Midterm Exam + Revision

Perplexity.AI

- Construct the Context-Free Grammar for the language where the number of $b + 2 =$ the number of a . The number of a is more than b but with strictly only two letters
 - $S \rightarrow aSb \mid b$
 - $D \rightarrow aDb \mid k$
 - ?

Construct the Context-Free Grammar for the language where the number of $b + 2 =$ the number of a . The number of a is more than b but with strictly only two letters

Pro Search

Sources 8

72 tasks ↗



Tutorialspoint

Introduction to Context-Free Grammar

Wikimedia Foundation, L...

Context-free language - Wikipedia



Tutorialspoint

What is context free grammar? Explain with examples

+5 sources

To construct a context-free grammar (CFG) for the language where the number of b 's plus 2 equals the number of a 's (i.e., $a = b + 2$), we need to ensure that every string generated maintains this balance. The grammar must allow a 's and b 's to be interleaved while preserving the required surplus of two a 's.

Approach

- Base Case:** Start with two a 's to establish the initial surplus.
- Recursive Rules:** For each b added, include two a 's to maintain the surplus. Allow these elements to be inserted anywhere in the string.

CFG Definition

The grammar is defined as:

text

$S \rightarrow aa \mid aSa \mid bSaa \mid baSa \mid baaS \mid aabS \mid abaS \mid abSa$

Midterm + Revision

- Context-Free Grammar

- Construct a context-free grammar for the language where the number of a's is more than the number of b's by exactly 2.

number of a's is more than the number of b's by exactly 2.

Construct a context-free grammar for the language where the number of a's is more than the number of b's by exactly 2.



- 1. Construct the Context-Free Grammar for the language where the number of b + 2 = the number of a. The number of a is more than b but with strictly only two letters. The a and b should be at any order ?

I'll construct a context-free grammar for a language where:

1. The number of 'a's equals the number of 'b's plus 2
2. The 'a's and 'b's can appear in any order

Claude.AI

Let's define this grammar:

$S \rightarrow aaS \mid aSa \mid Saa \mid aSb \mid bSa \mid abS \mid baS \mid Sab \mid Sba \mid aabb \mid abab \mid abba \mid baab \mid baba \mid bbaa$

Let me explain this grammar:

- The start symbol S generates strings where $\#a = \#b + 2$
- Each production rule maintains this property
- The last six options (aabb, abab, etc.) are the base cases with 2 'a's and 2 'b's

However, this approach is overly complex. Let me provide a more elegant solution:

$S \rightarrow aaSb \mid abSa \mid baSa \mid bbSaa \mid \epsilon$

Midterm

+ Problem

- ! Construct the Context-Free Grammar for the language where the number of $b + 2 =$ the number of a . The number of a is more than b but with strictly only two letters. The a and b should be at any order?

e.AI

aa

Don't ever use Generative AI tools for this course

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However, this approach is overly complex. Let me provide a more elegant solution:

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Midterm Exam Questions + Revision

- **Context Free Grammars**

- Construct the Context-Free Grammar for the language where the number of b + 2 = the number of a . The number of a is more than b but with strictly only two letters.

- $S \rightarrow aSb \mid bSa \mid SDD \mid DSD \mid DDS \mid aa \mid ?$

- $D \rightarrow aDb \mid bDa \mid DD \mid e$

- ?

Midterm Exam Questions + Revision

- **Context Free Grammars**

- Construct the Context-Free Grammar for the language where the number of b + 2 = the number of a . The number of a is more than b but with strictly only two letters.

- $S \rightarrow aSb \mid bSa \mid SDD \mid DSD \mid DDS \mid aa \ ?$

- $D \rightarrow aDb \mid bDa \mid DD \mid e$

- **abaa ?**

Midterm Exam Questions + Revision

- **Context Free Grammars**

- Construct the Context-Free Grammar for the language where the number of b + 2 = the number of a . The number of a is more than b but with strictly only two letters.

- $S \rightarrow DaDaD$
- $D \rightarrow aDb \mid bDa \mid DD \mid e$
 - **abaa ?**

Chomsky Classification of Languages

- We have moved to study Context-Free Languages and Grammar

Type	Grammar	Language	Automaton
Type-3	Regular Grammar	Regular Languages	DFA/NFA
Type-2	Context-Free Grammar	Context Free Languages	PDA

Chomsky Classification of Languages

- What about Context-sensitive languages ?

Type	Grammar	Language	Automaton
Type-3	Regular Grammar	Regular Languages	DFA/NFA
Type-2	Context-Free Grammar	Context Free Languages	PDA
Type-1	Context-Sensitive Grammar	Context-Sensitive Languages	Linear-bounded automaton

Chomsky Classification of Languages



- We have started with Regular languages

Type	Grammar	Language	Automaton
Type-3	Regular Grammar	Regular Languages	DFA/NFA

Chomsky Classification of Languages

- What about Context-sensitive languages ?

- Example of Grammars

- $S \rightarrow abc \mid aAbc$
- $Ab \rightarrow bA$
- $Ac \rightarrow Bbcc$
- $bB \rightarrow Bb$
- $aB \rightarrow aa \mid aaA$

**What language is
for this grammar ?**

Chomsky Classification of Languages

- What about Context-sensitive languages ?

- Example of Grammars

- $S \rightarrow abc \mid aAbc$
- $Ab \rightarrow bA$
- $Ac \rightarrow Bbcc$
- $bB \rightarrow Bb$
- $aB \rightarrow aa \mid aaA$

$S \rightarrow aAbc$

$\rightarrow abAc$

$\rightarrow abBbcc$

$\rightarrow aBbbcc$

$\rightarrow aaAbbcc$

$\rightarrow aabAbcc$

$\rightarrow aabbAcc$

$\rightarrow aabbBbcc$

$\rightarrow aabBbbcc$

$\rightarrow aaBbbbcc$

$\rightarrow \mathbf{aaabbbccc}$

Chomsky Classification of Languages

- Lastly, Turing Machine will be explained next week

Type	Grammar	Language	Automaton
Type-3	Regular Grammar	Regular Languages	DFA/NFA
Type-2	Context-Free Grammar	Context Free Languages	PDA
Type-1	Context-Sensitive Grammar	Context-Sensitive Languages	Linear-bounded automaton
Type-0	Unrestricted grammar	Recursively enumerable language	Turing Machine

Chomsky Classification of Languages



- Context-free languages are those that can be **generated** by context-free grammar.
- Example : Language $L = \{ 0^n 1^n \mid n \geq 0 \}$

Pumping Lemma for Context Free Grammar



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- Example : Language $L = \{ 0^n 1^n \mid n \geq 0 \}$

Pumping Lemma for Context Free Grammar



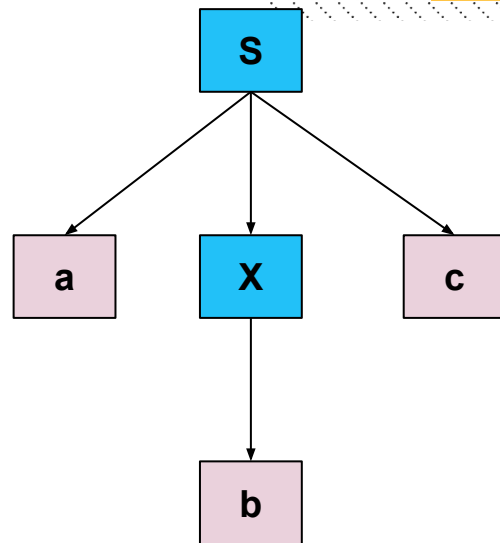
- Context-free languages are those that can be **generated** by context-free grammar.
- What about : Language $L = \{ a^n b^n c^n \mid n \geq 0 \}$?

Pumping Lemma for Context Free Grammar

- Given the following grammar:
 - $S \rightarrow aX$
 - $X \rightarrow bc$
- Let's derive the string **abc**

Pumping Lemma for Context Free Grammar

- Given the following grammar:
 - $S \rightarrow aXc$
 - $X \rightarrow b$
- Let's derive the string **abc**



Pumping Lemma for Context Free Grammar

- Given the following grammar:

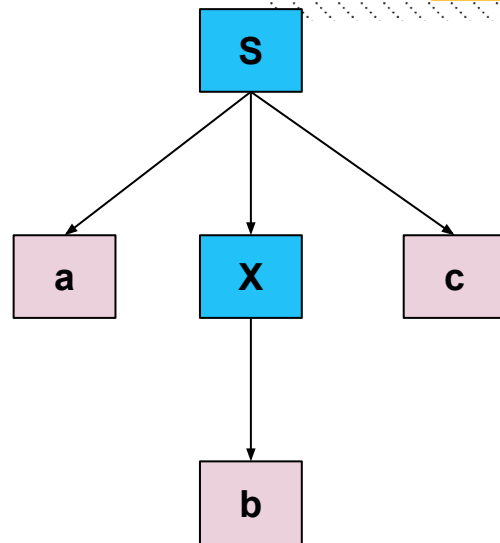
- $S \rightarrow aXc$

- $X \rightarrow b$

- Can we generate:

- More or infinite number of words ?

- Longer** words ?



Pumping Lemma for Context Free Grammar

We can only Generate a single word

- Given the following grammar:

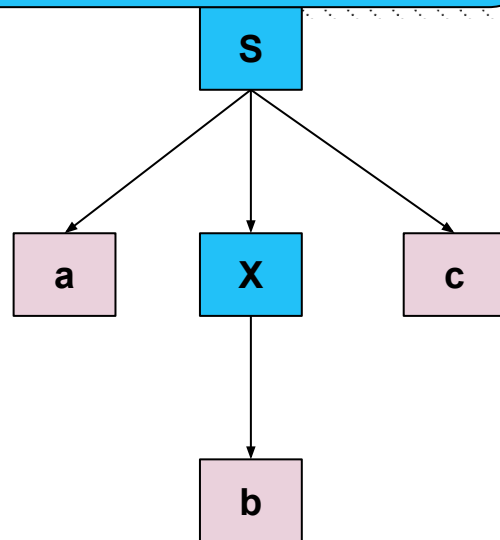
- $S \rightarrow aXc$

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Pumping Lemma for Context Free Grammar

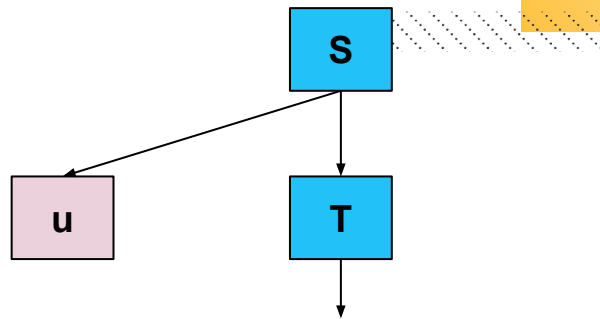
- Let's consider the grammar :

- $S \rightarrow uT \mid \varepsilon$

- $T \rightarrow vT \mid x \mid \varepsilon$

- Can we generate:

- ???



Pumping Lemma for Context Free Grammar

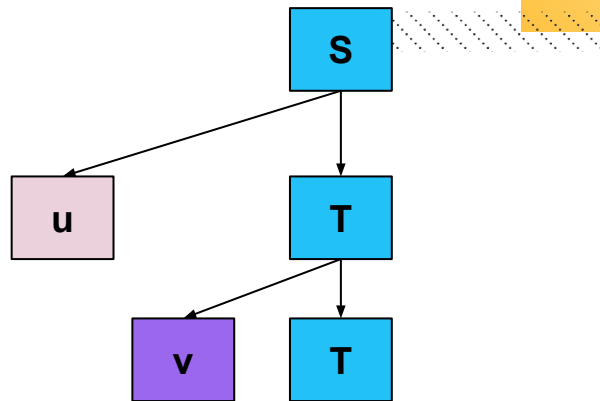
- Let's consider the grammar :

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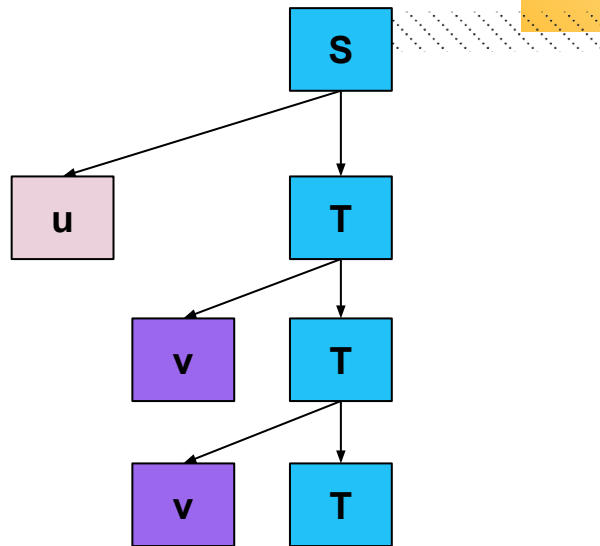
- Can we generate:

- $u\mathbf{vvv}x$, $u\mathbf{vvvvvvvv}x$, $u\mathbf{vvvvvvvv}\dots\mathbf{vv}x$



Pumping Lemma for Context Free Grammar

- Let's consider the grammar :
 - $S \rightarrow uT \mid \epsilon$
 - $T \rightarrow vT \mid x \mid \epsilon$
- Can we generate:
 - $u\mathbf{vvv}x$, $u\mathbf{vvvvvvvv}x$, $u\mathbf{vvvvvvvv}\dots\mathbf{vv}x$

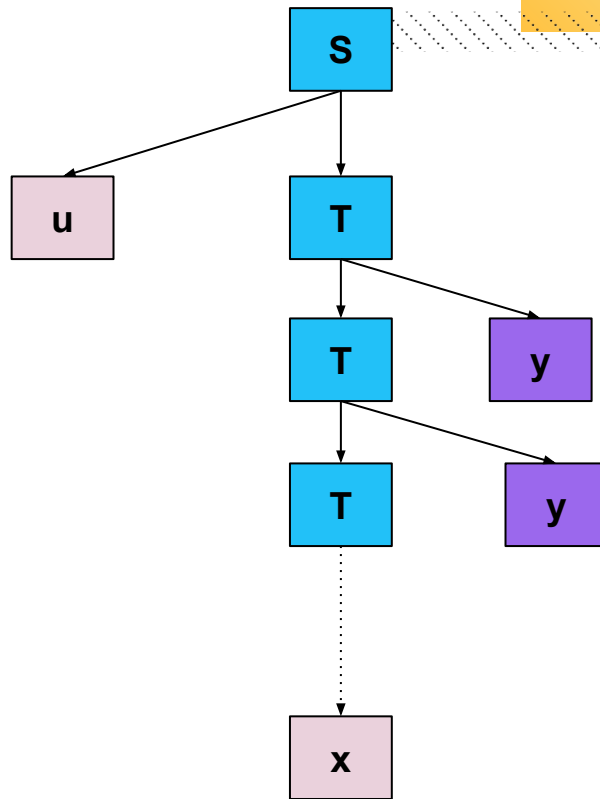


Pumping Lemma for Context Free Grammar

- Let's consider the grammar :
 - $S \rightarrow uT \mid \varepsilon$
 - $T \rightarrow T\mathbf{y} \mid x \mid \varepsilon$
- Can we generate:
 - $ux\mathbf{yy}$, $ux\mathbf{yyyyyyy}$, $ux\mathbf{yyyyyyy...yy}$

Pumping Lemma for Context Free Grammar

- Let's consider the grammar :
 - $S \rightarrow uT \mid \varepsilon$
 - $T \rightarrow T\mathbf{y} \mid x \mid \varepsilon$
- Can we generate:
- $ux\mathbf{yy}$, $ux\mathbf{yyyyyyy}$, $ux\mathbf{yyyyyyy...yy}$



Pumping Lemma for Context Free Grammar

- Let's consider the grammar :
 - $S \rightarrow uTz \mid \varepsilon$
 - $T \rightarrow \mathbf{vTy} \mid x \mid \varepsilon$
- Can we generate:
 - $u\mathbf{vvv}x\mathbf{yyy}z$, $u\mathbf{vvvvvvvv}x\mathbf{yyyyyy}z$, $u\mathbf{vvvvvvvv}\dots\mathbf{vv}x\mathbf{yyyyyy}\dots\mathbf{yy}z$

Pumping Lemma for Context Free Grammar

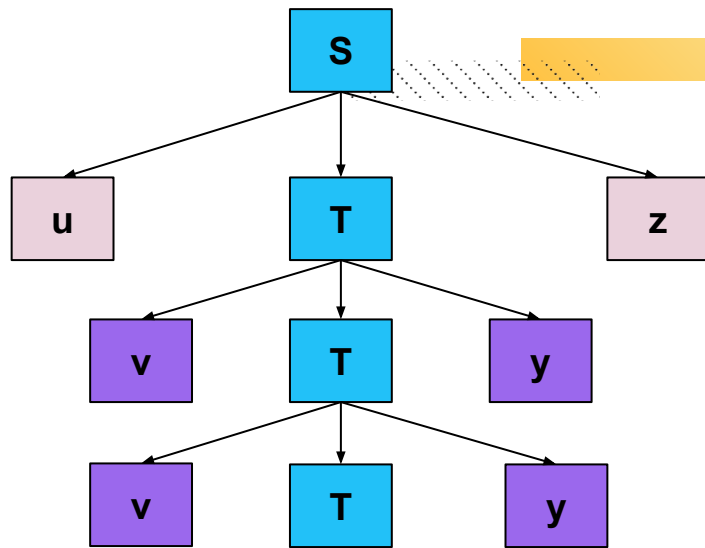
- Let's consider the grammar :

- $S \rightarrow uTz \mid \epsilon$

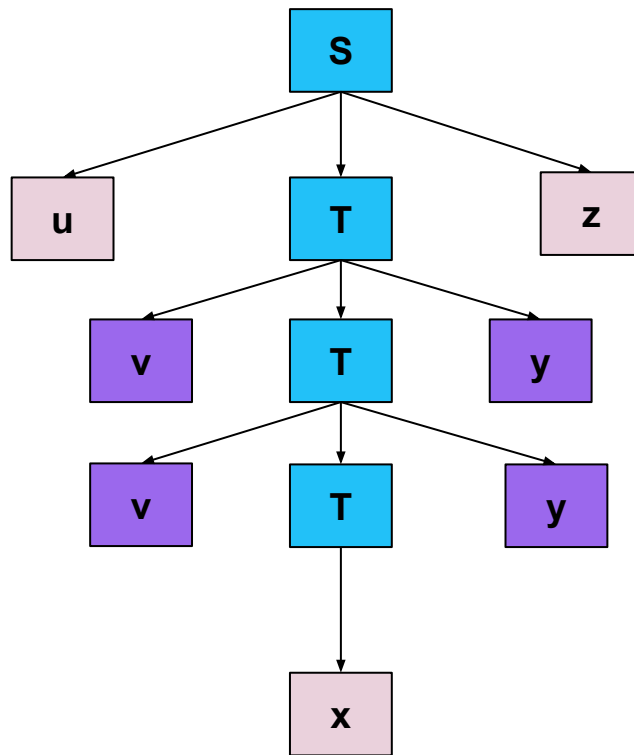
- $T \rightarrow vTy \mid x \mid \epsilon$

- Can we generate:

- $uvvvxyyyz$, $uvvvvvvvxyyyyyyz$, $uvvvvvvv...vvxyyyyyy...yyz$

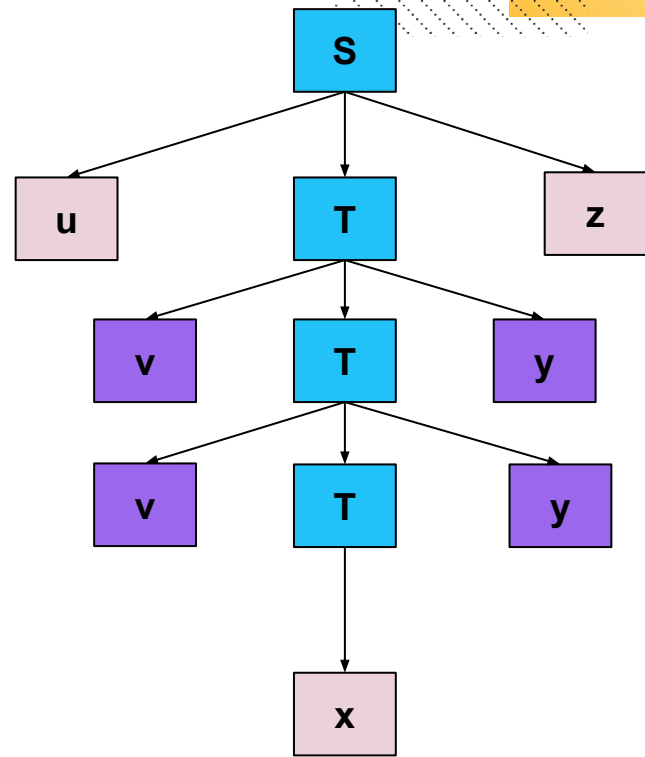


Pumping Lemma for Context Free Grammar



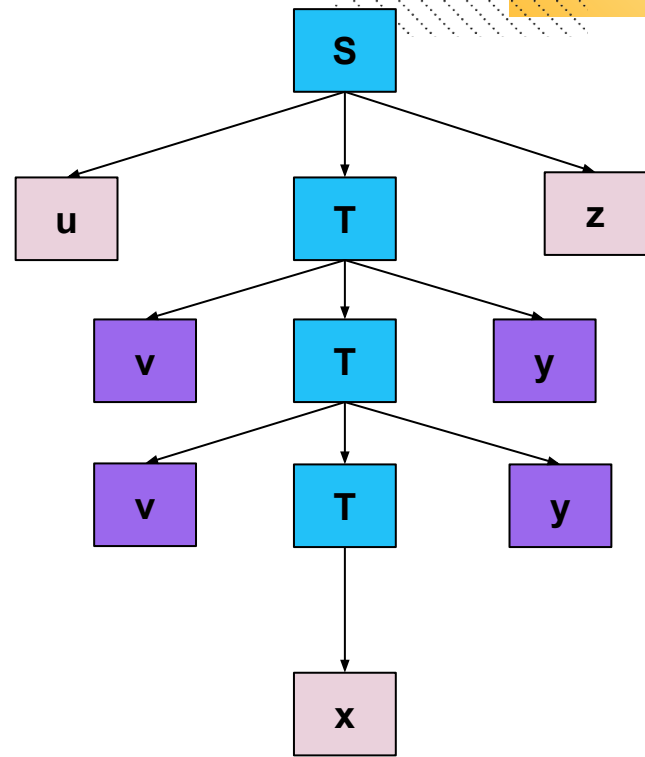
Pumping Lemma for Context Free Grammar

- For a given string belonging to the language which is infinite, it must be in the form : $uvxyz$
- Such that :



Pumping Lemma for Context Free Grammar

- For a given string belonging to the language which is infinite, it must be in the form : $uvxyz$
- Such that : **pumping can happen for v or y or both**



Pumping Lemma for Context Free Grammar

Pumping lemma for context-free languages If A is a context-free language, then there is a number p (the pumping length) where, if s is any string in A of length at least p , then s may be divided into five pieces $s = uvxyz$ satisfying the conditions

1. for each $i \geq 0$, $uv^i xy^i z \in A$,
2. $|vy| > 0$, and
3. $|vxy| \leq p$.

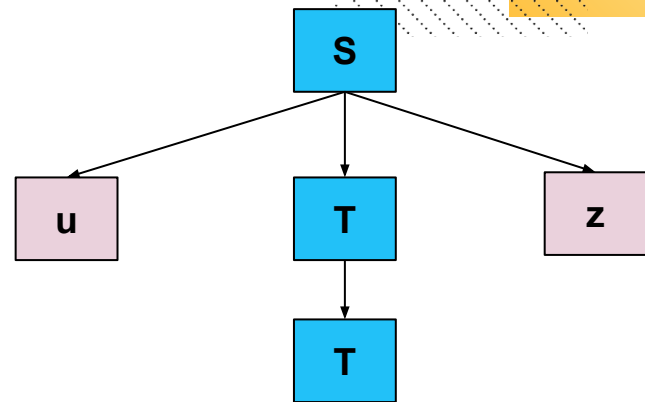


Pumping Lemma for Context Free Grammar

- What if $|vy| = 0$?
 - $S \rightarrow uTz \mid \epsilon$
 - $T \rightarrow \mathbf{vTy} \mid x \mid \epsilon$
- Would be :
 - $S \rightarrow uTz \mid \epsilon$
 - $T \rightarrow T \mid x \mid \epsilon$

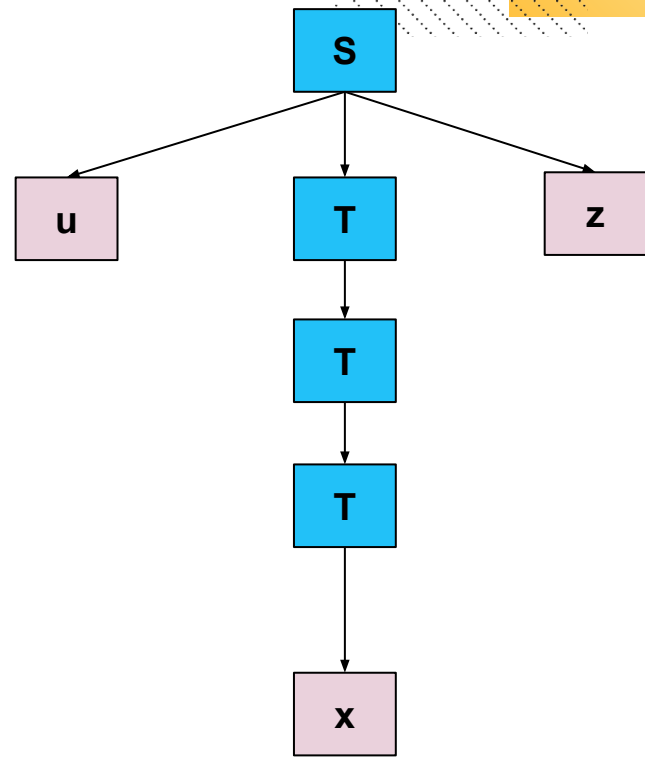
Pumping Lemma for Context Free Grammar

- What if $|vy| = 0$?
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- Would be :
 - $S \rightarrow uTz \mid \epsilon$
 - $T \rightarrow T \mid x \mid \epsilon$



Pumping Lemma for Context Free Grammar

- What if $|vy| = 0$?
 - $S \rightarrow uTz \mid \varepsilon$
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- Would be :
 - $S \rightarrow uTz \mid \varepsilon$
 - $T \rightarrow T \mid x \mid \varepsilon$



Pumping Lemma for Context Free Grammar

- We wish to show

-
-
-

- We

- $S \rightarrow uTz \mid \epsilon$

- $T \rightarrow T \mid x \mid \epsilon$

No matter you how many recursive substitutions you do, you end up having the same word : uxz

S

z

x

Pumping Lemma for Context Free Grammar

Pumping lemma for context-free languages If A is a context-free language, then there is a number p (the pumping length) where, if s is any string in A of length at least p , then s may be divided into five pieces $s = uvxyz$ satisfying the conditions

1. for each $i \geq 0$, $uv^ixy^iz \in A$,
2. $|vy| > 0$, and
3. $|vxy| \leq p$.

There is another technique called Ogden's lemma

Pumping Lemma for Context Free Grammar

- If you cannot for a given language :
 - Create the Context-Free Grammar
 - Create a Pushdown automaton
- You may use the pumping lemma to prove it is not a context-free language

Pumping Lemma for Context Free Grammar

- To use the pumping lemma via Proof by Contradiction:
 - Assume the language is a context free language
 - Assume the pumping length **P**
 - Think of a string which is part of the language , such that if you put it in the form : $u\mathbf{v}xyz$,
 - Regardless how you choose **v** and **y** from the string , pumping them, would lead to generate strings not belonging to the language

Pumping Lemma for Context Free Grammar



- Use the pumping lemma to show that the language $B = \{a^n b^n c^n \mid n \geq 0\}$ is not context free

Pumping Lemma for Context Free Grammar

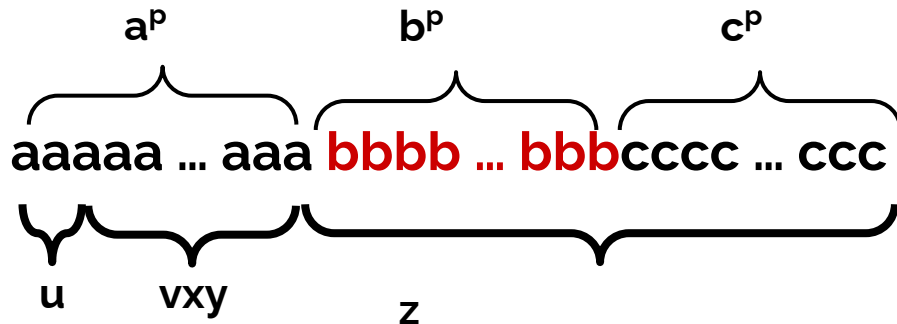
- Use the pumping lemma to show that the language $B = \{a^n b^n c^n \mid n \geq 0\}$ is not context free
 - We assume it is context-free.
 - We assume the pumping length P
 - Let's try this string : $a^P b^P c^P$

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $B = \{a^n b^n c^n \mid n \geq 0\}$ is not context free
 - String : $a^P b^P c^P \Rightarrow uvxyz$ such that : $|vxy| \leq P$
 - Let's try all possible combinations for placing the **vxy** on our string $a^P b^P c^P$ and see if we can pump

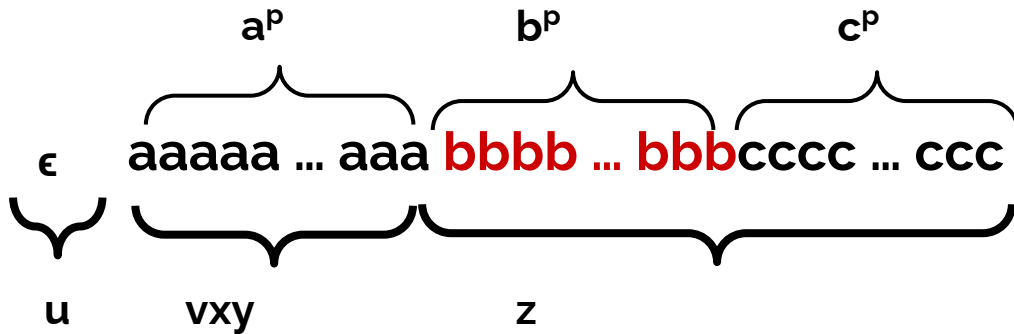
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- Use the pumping lemma to show that the language $B = \{a^n b^n c^n \mid n \geq 0\}$ is not context free



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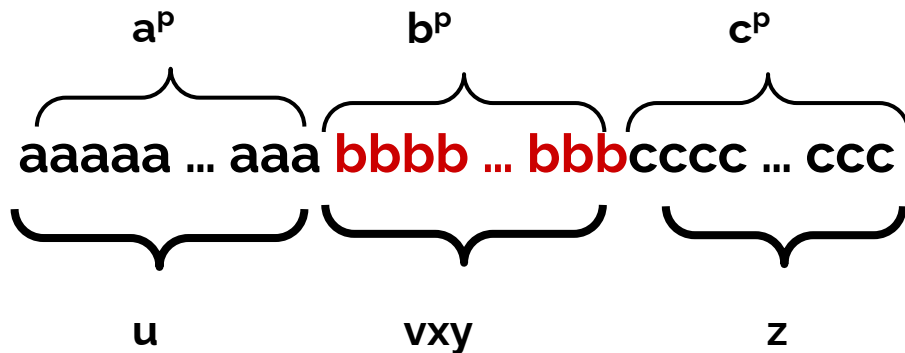


Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $B = \{a^n b^n c^n \mid n \geq 0\}$ is not context free
 - String : $a^P b^P c^P \Rightarrow uvxyz$ such that : $|vxy| \leq P$
 - At first part : $a^P \Rightarrow$ **you can pump just a ? generated words**
would have more a than b and c $\Rightarrow$ not in the language

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $B = \{a^n b^n c^n \mid n \geq 0\}$ is not context free

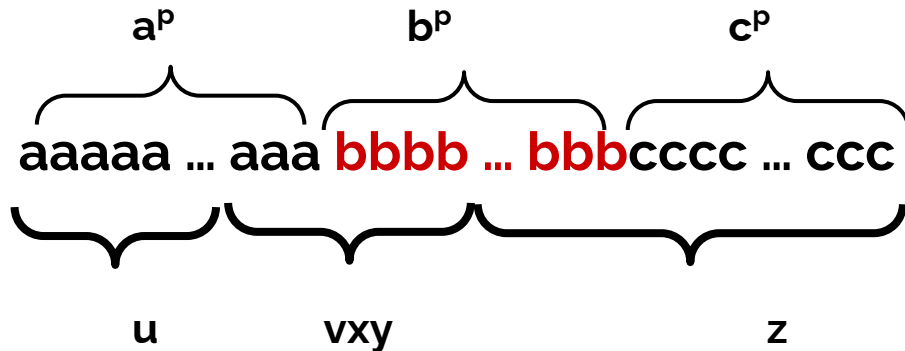


Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $B = \{a^n b^n c^n \mid n \geq 0\}$ is not context free
 - String : $a^P b^P c^P \Rightarrow uvxyz$ such that : $|vxy| \leq P$
 - At middle part between a and b ?

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $B = \{a^n b^n c^n \mid n \geq 0\}$ is not context free



Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $B = \{a^n b^n c^n \mid n \geq 0\}$

is not context free

- String : $a^P b^P c^P \Rightarrow uvxyz$ such that : $|vxy| \leq P$

- At middle part between a and b :

You will pump only a and b but not C.

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $B = \{a^n b^n c^n \mid n \geq 0\}$ is not context free
 - String : $a^P b^P c^P \Rightarrow uvxyz$ such that : $|vxy| \leq P$
 - At middle part : Between b and c : Same

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $B = \{a^n b^n c^n \mid n \geq 0\}$ is not context free
 - String : $a^P b^P c^P \Rightarrow uvxyz$ such that : $|vxy| \leq P$
 - Between a^P and $b^P \Rightarrow$ **regardless of what you take for v and y , the c will not be pumped.**

Pumping Lemma for Context Free Grammar

- Language :

WordWord

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $D = \{ww \mid w \in \{0,1\}^*\}$ is not context free
 - We assume it is context-free.
 - We assume the pumping length **P**
 - Let's try this string : ?

Pumping Lemma for Context Free Grammar

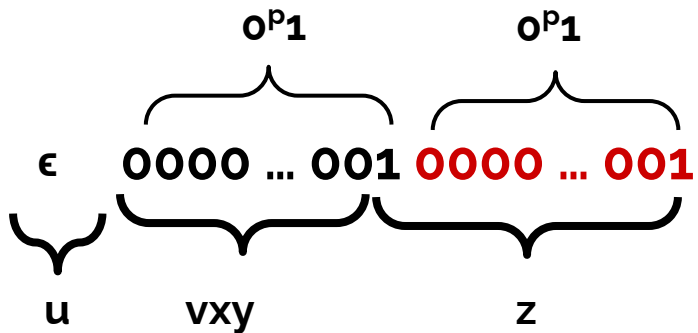
- Use the pumping lemma to show that the language $D = \{ww \mid w \in \{0,1\}^*\}$ is not context free
 - We assume it is context-free.
 - We assume the pumping length P
 - Let's try this string : $0^P 1 0^P 1$

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $D = \{ww \mid w \in \{0,1\}^*\}$ is not context free
 - String : $0^P 1 0^P 1 \Rightarrow uvxyz$ such that : $|vxy| \leq P$
 - Let's try all possible combinations for placing the $uvxyz$ on our string $0^P 1 0^P 1$ **and see if we can pump**

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $D = \{ww \mid w \in \{0,1\}^*\}$ is not context free



Pumping Lemma for Context Free Grammar

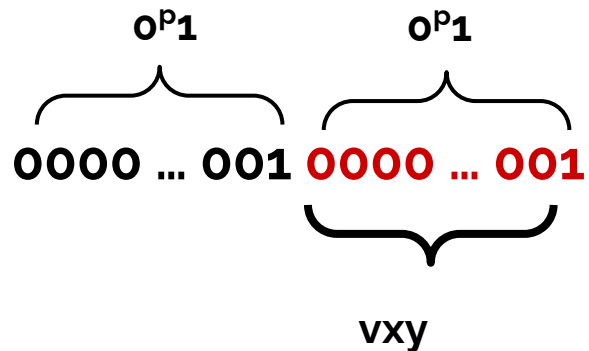
- Use the pumping lemma to show that the language $D = \{ww \mid w \in \{0,1\}^*\}$ is not context free

$$\underbrace{0000 \dots 001}_{vxy} \quad \underbrace{0000 \dots 001}_{0^p 1}$$

The second word would not be equal to the first word

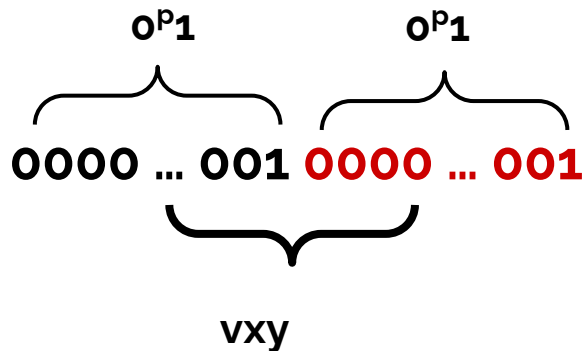
Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $D = \{ww \mid w \in \{0,1\}^*\}$ is not context free



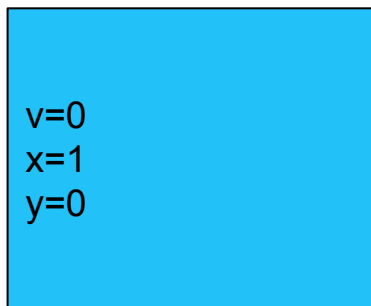
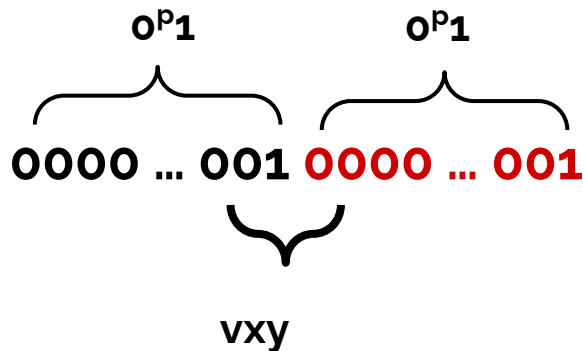
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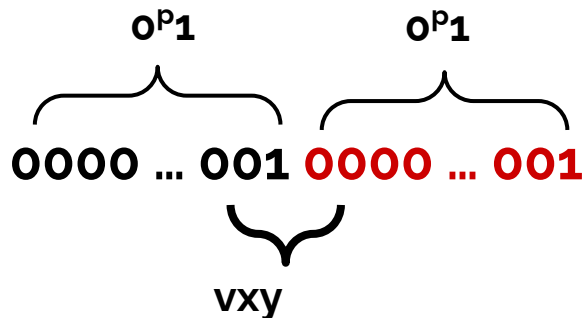
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Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $D = \{ww \mid w \in \{0,1\}^*\}$ is not context free



$v=0$
 $x=1$
 $y=0$

- We can pump the selected string and end up generating words in the same language

Pumping Lemma for Context Free Grammar

- $U \in \{0,1\}^*$

This does not mean that the language is context-free, it means you chose a bad string

vxy

$x=1$
 $y=0$

- We can pump the selected string and ends up generating words in the same language

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $D = \{ww \mid w \in \{0,1\}^*\}$ is not context free
 - String : $0^P 1 0^P 1 \Rightarrow uvxyz$ such that : $|vxy| \leq P$
 - Let's try all possible combinations for placing the $uvxyz$ on our string $0^P 1^P 0^P 1^P$ **and see if we can pump**

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $D = \{ww \mid w \in \{0,1\}^*\}$ is not context free

$0^{p1^p} \quad 0^{p1^p}$

$0000 \dots 001111\dots111 \quad 0000 \dots 001111\dots111$

vxy

More ones in the first word + if you break to get two words ? $\Rightarrow$ second word starts with 1, not zero

Pumping Lemma for Context Free Grammar

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More numbers added ? if you break to get two word ? $\Rightarrow$ second word starts with 1, not zero

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- Use the pumping lemma to show that the language $D = \{ww \mid w \in \{0,1\}^*\}$ is not context free

$0^{p1^p} \quad 0^{p1^p}$

$0000 \dots 001111 \dots 111 \quad 0000 \dots 001111 \dots 111$

vxy

Even if you take x as empty string ...

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $C = \{a^i b^j c^k \mid 0 \leq i \leq j \leq k\}$ is not context free
 - We assume it is context-free.
 - We assume the pumping length P
 - Let's try this string : $a^P b^P c^P$

Pumping Lemma for Context Free Grammar

Pumping lemma for context-free languages If A is a context-free language, then there is a number p (the pumping length) where, if s is any string in A of length at least p , then s may be divided into five pieces $s = uvxyz$ satisfying the conditions

1. for each $i \geq 0$, $uv^i xy^i z \in A$,
2. $|vy| > 0$, and
3. $|vxy| \leq p$.

Pumping Lemma for Context Free Grammar

Pumping lemma for context-free languages If A is a context-free language, then there is a number p (the pumping length) where, if s is any string in A of length at least p , then s may be divided into five pieces $s = uvxyz$ satisfying the conditions

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Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $C = \{a^i b^j c^k \mid 0 \leq i \leq j \leq k\}$ is not context free
 - Example of words:
 - abbccc
 - bc
 - abcc

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $C = \{a^i b^j c^k \mid 0 \leq i \leq j \leq k\}$ is not context free

$a^p b^p c^p$

aaaa...aaaa **bbbb** ...**bbbb** cccc... ccccc

vxy

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $C = \{a^i b^j c^k \mid 0 \leq i \leq j \leq k\}$ is not context free

$a^p b^p c^p$

aaaa...aaaa **bbbb** ...**bbbb** cccc... ccccc

vxy

Seems Easy ?
We just pump **up**
v and y which can
be c ?

**Words will be
Always in the
language**

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $C = \{a^i b^j c^k \mid 0 \leq i \leq j \leq k\}$ is not context free

$a^p b^p c^p$

aaaa...aaaa **bbbb** ...**bbbb** cccc... ccccc

vxy

Can we design the PDA ?

Seems Easy ?
We just pump **up**
v and y which can
be c ?

**Words will be
Always in the
language**

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $C = \{a^i b^j c^k \mid 0 \leq i \leq j \leq k\}$ is not context free

$a^p b^p c^p$

aaaa...aaaa **bbbb** ...**bbbb** cccc... ccccc

vxy

v or y contains only c

If we pump **UP** ?

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $C = \{a^i b^j c^k \mid 0 \leq i \leq j \leq k\}$ is not context free

$a^p b^p c^p$

aaaa...aaaa **bbbb** ...**bbbb** cccc... ccccc

vxy

v or y contains only c
Let's assume $i=0$ for
the pumping lemma
“Pumping down”

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $C = \{a^i b^j c^k \mid 0 \leq i \leq j \leq k\}$ is not context free

$a^p b^p c^p$

aaaa...aaaa **bbbb** ...**bbbb** cccc... ccccc

vxy

$v^0xy^0 \Rightarrow$ Certainly, we will drop at least one C

Remember $|vy| > 1$

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $C = \{a^i b^j c^k \mid 0 \leq i \leq j \leq k\}$ is not context free

$a^p b^p c^p$

aaaa...aaaa **bbbb** ...**bbbb** cccc... ccccc

vxy

$uv^0xy^0z \Rightarrow$ does it belong to the language ?

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $C = \{a^i b^j c^k \mid 0 \leq i \leq j \leq k\}$ is not context free

$a^p b^p c^p$

aaaa...aaaa **bbbb** ...**bbbb** cccc... ccccc

vxy

$uv^0xy^0z \Rightarrow$ does it belong to the language ?
No because the number of C is less than P

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $C = \{a^i b^j c^k \mid 0 \leq i \leq j \leq k\}$ is not context free

$a^p b^p c^p$

aaaa...aaaa **bbbb** ...**bbbb** cccc... ccccc

vxy

Same case as before,
number of A should
 $\leq B$ or C

Pumping Lemma for Context Free Grammar

- Use the pumping lemma to show that the language $C = \{a^i b^j c^k \mid 0 \leq i \leq j \leq k\}$ is not context free

$a^p b^p c^p$

aaaa...aaaa **bbbb** ...**bbbb** cccc... ccccc

vxy

Pump down or pump up ?

Non-Regular Languages

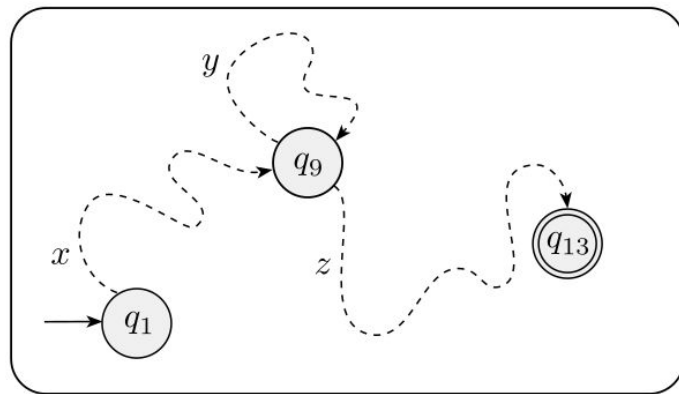
Pumping Lemma

- **Theorem**

- Pumping lemma If A is a regular language, then there is a number p (the pumping length) where if s is any string in A of length **at least** p , then s may be divided into three pieces, $s = xyz$, satisfying the following conditions:

- for each $i \geq 0$, $xy^i z \in A$,
- $|y| > 0$, and
- $|xy| \leq p$.

- p is called the **pumping length**



Closure Properties for Context Free Grammar

- If L_1 and L_2 are context-free languages over an alphabet Σ , then
 - $L_1 \cup L_2$
 - $L_1 L_2$
 - L_1^*
- are also Context-free languages.

Closure Properties for Context Free Grammar

- **Union**

- Given two languages:
 - $G_1 = (N_1, \Sigma, S_1, P_1)$ be CFG for L_1 .
 - $G_2 = (N_2, \Sigma, S_2, P_2)$ be CFG for L_2
- $G_u = G_1 \cup G_2$ where $G_u = (N_u, \Sigma, S_u, P_u)$
 - $N_u = N_1 \cup N_2 \cup \{S_u\}$
 - $P_u = P_1 \cup P_2 \cup \{S_u \rightarrow S_1 \mid S_2\}$

N : non-terminal variables
P : Production rules
S : Start Variable.

Closure Properties for Context Free Grammar



- Union : Important
 - If L_1 and L_2 are context Free Languages $\Rightarrow L_1 \cup L_2$ is context free language.
 - If L_1 and $L_1 \cup L_2$ are context free languages $\Rightarrow$ does it imply that L_2 is also context free language ?

Closure Properties for Context Free Grammar

- Union : Important
 - Consider : $L_1 = \Sigma^*$
 - Consider L_2 any non context free language ($a^n b^n c^n$ where $n > 0$)
 - $L_1 \cup L_2 = L_1$

Closure Properties for Context Free Grammar

- **Union : Important**

- If L_1 and L_2 are context Free Languages $\Rightarrow L_1 \cup L_2$ is context free language.
- If L_1 and $L_1 \cup L_2$ are context free languages $\Rightarrow$ does it imply that L_2 is also context free language ?

■ **Not necessarily**

Closure Properties for Context Free Grammar

- **Concatenation**

- Given two languages:
 - $G_1 = (N_1, \Sigma, S_1, P_1)$ be CFG for L_1 .
 - $G_2 = (N_2, \Sigma, S_2, P_2)$ be CFG for L_2
- $G_c = G_1 G_2$ where $G_c = (N_c, \Sigma, S_c, P_c)$
 - $N_c = N_1 \cup N_2 \cup \{S_c\}$
 - $P_c = P_1 \cup P_2 \cup \{S_c \rightarrow S_1 S_2\}$

Closure Properties for Context Free Grammar

- **Star**

- Given the following language:
 - $G_1 = (N_1, \Sigma, S_1, P_1)$ be CFG for L_1 .
- G_s is the grammar for L^* where $G_s = (N_s, \Sigma, S_s, P_s)$
 - $N_s = N_1 \cup \{S_s\}$
 - $P_s = P_1 \cup \{S_s \rightarrow S_1 S_s \mid \epsilon\}$

Closure Properties for Context Free Grammar

- Intersection is not closed for context-free grammar :
 - Given the two context free languages languages:
 - $L_1 = \{a^n b^n c^k \mid n \text{ and } k \geq 0\}$
 - $L_2 = \{a^k b^n c^n \mid n \text{ and } k \geq 0\}$
 - $L_1 \cap L_2 = ?$

Closure Properties for Context Free Grammar

- Intersection is not closed for context-free grammar :
 - Given the two context free languages languages:
 - $L_1 = \{a^n b^n c^k \mid n \text{ and } k \geq 0\}$
 - $L_2 = \{a^k b^n c^n \mid n \text{ and } k \geq 0\}$
 - $L_1 \cap L_2 = \{a^n b^n c^n \mid n \geq 0\}$ this is non context free grammar

Closure Properties for Context Free Grammar



- Complement is not closed under the Context-Free Languages
 - Example :
 - Language $\{ wx \text{ such that } w \text{ is not } x \}$
 - Can we design DFA/NFA/PDA for it ? or the CFG ?
 - **No ? Yes ?**